

Best model chosen

Relevant figures and diagrams

Confusion Matrix:

```
[[ 0  0  0  0  0  0  0  0  0  0  0]
 [ 9 21  1  0  0 153  1  0 159  0 29]
 [ 5  0  3  0  0 90  0 31 14 98  0]
 [40  0  0 122  0 366 17  0 171  0 489]
 [18  0  0  0 35  0  0  0  0  0  0]
 [68  0  0  0  0 113  0 11 51  2  0]
 [ 4  1  0  3  0 40 233  0 42  0 223]
 [ 3  0  0  0  0  0  0 21  0  0  0]
 [21  0  0  0  0  0  0  2 53  0  0]
 [ 5  0  0  0 16  0  0  5  0 100  0]
 [ 9  0  0  0  0  0  0  0 20  0 115]]
```

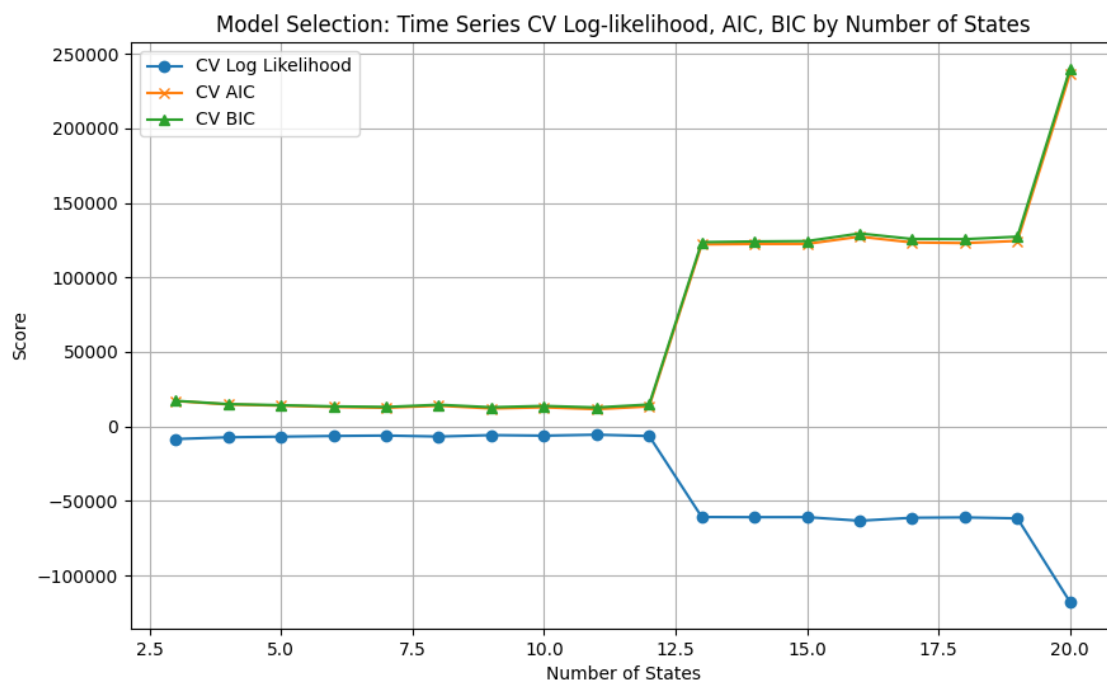
Weighted Average Classification Report:

```
{'precision': 0.7879553180628392, 'recall': 0.26904055390702275, 'f1-score': 0.2675312452338454}
```

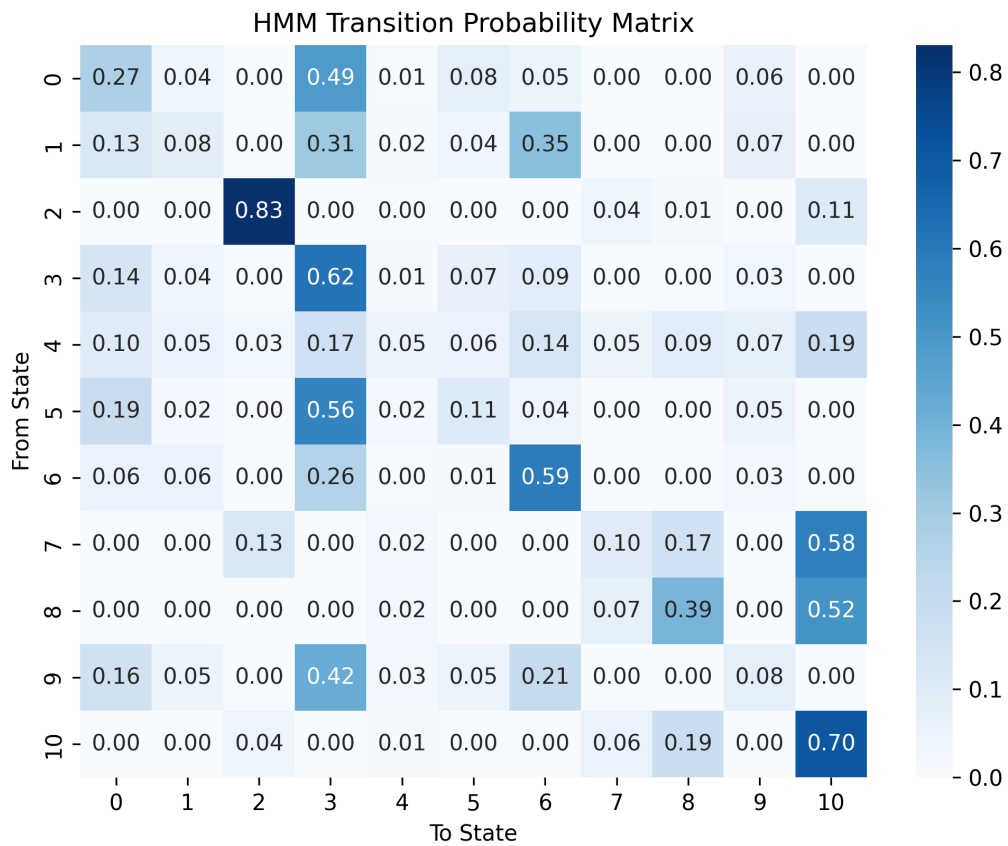
best model chosen metrics:

- Best model selected: 11 states
- log likelihood: 3112.134422166114,
- aic: -5186.268844332228,
- bic: -2039.8662255005493

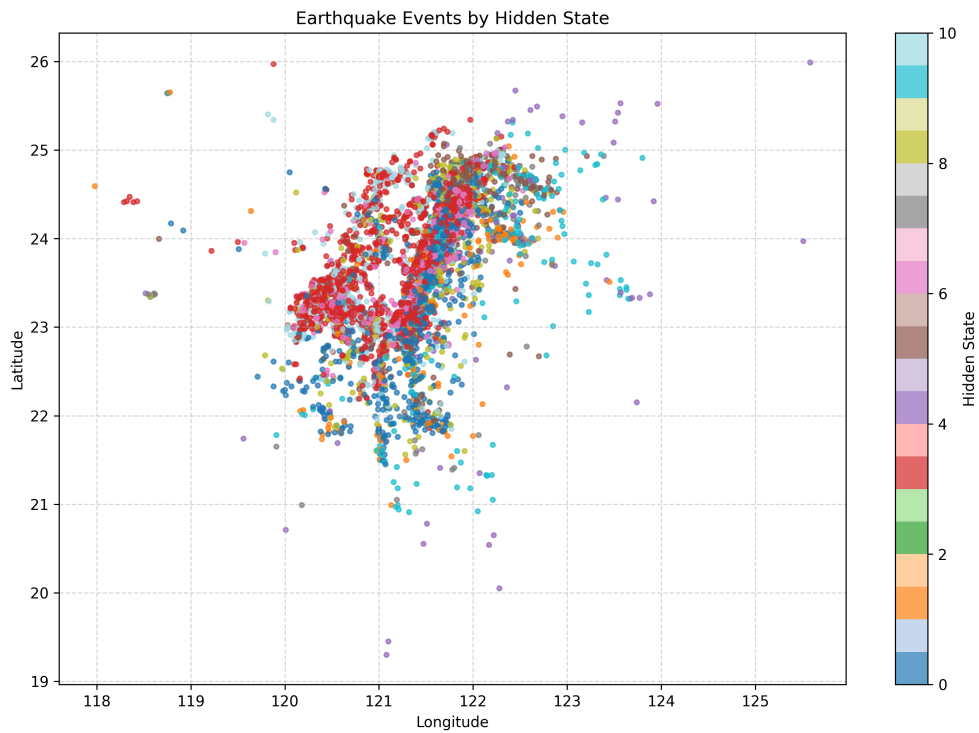
With the plots, we can see that after state 11, the states were not converging:



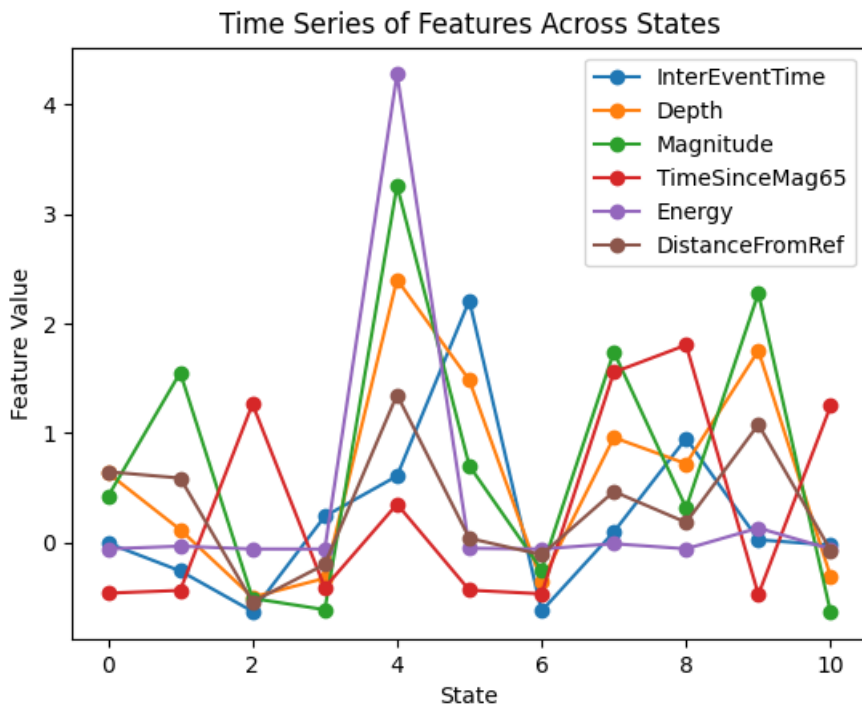
We can see that the matrix is quite sparse:



The model becomes a bit hard to interpret when looking geographically:



So, looking at the features we get:



Analysis

Transition matrix

- Diagonal dominance in transition matrix: Strong self-transition probabilities suggest states tend to persist, indicating earthquake activity has temporal clustering or "stickiness"
 - the aftershocks
- Limited State Connectivity: Most states only transition to 2-3 other states, suggesting distinct seismic regimes with restricted pathways between them
- Potential Absorbing States: Some states may act as temporary "attractors" in the seismic cycle?
 - Like state 10

Geographic interpretation

- There is a lot of spatial mixing,
- could be that states are more clustered due to temporal or magnitude based events rather than graphically (not clustered near fault line)
 - maybe state 6 could be related with severe earthquake
- more symptomatic of the univariate distribution with counts and no other features
- Different earthquake "phases" or "regimes" can occur across the entire study region

Feature state relation

The following is my interpretation from reading about how earthquakes work and trying to interpret the graph, use it as reference with caution ⚠️

States 3-6: Moderate Severe Earthquake Phase

- **Magnitude:** Elevated magnitude levels (4.5-5.5 range)
- **Inter-event Time:** Moderate to high inter-event times, suggesting these occur during building stress phases
- **Depth:** Mixed depth distribution, indicating both shallow and deeper moderate events
- **Time Since Mag 6.5:** Moderate values, representing intermediate phases in seismic cycles
- **Energy:** Elevated energy release corresponding to moderate-severe events
- **Distance from Reference:** Distributed across various distances from reference point

Seismic Interpretation: These states capture the transitional phase between background seismicity and major events - likely representing foreshock sequences, moderate mainshocks, or stress redistribution following major earthquakes.

States 8-10: Major Earthquake/Immediate Aftermath Phase

- **Magnitude:** Highest magnitude values (5.5+ range), representing major seismic events
- **Inter-event Time:** Very low values, indicating rapid succession of events (classic aftershock pattern)
- **Depth:** Relatively shallow depths, making these events particularly hazardous
- **Time Since Mag 6.5/5:** Very low values, confirming these states occur immediately following or during major earthquake sequences
- **Energy:** Peak energy release values, representing maximum seismic energy discharge
- **Distance from Reference:** Concentrated patterns suggesting spatial clustering of major events

Seismic Interpretation: These states represent the most critical seismic phases - major mainshocks (State 10) and their immediate aftershock sequences (States 8-9), characterized by high magnitude, shallow depth, and temporal clustering.